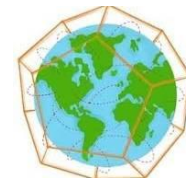


From Alligators to All In - Generalized Predator-Prey Problems and Guided Online Research

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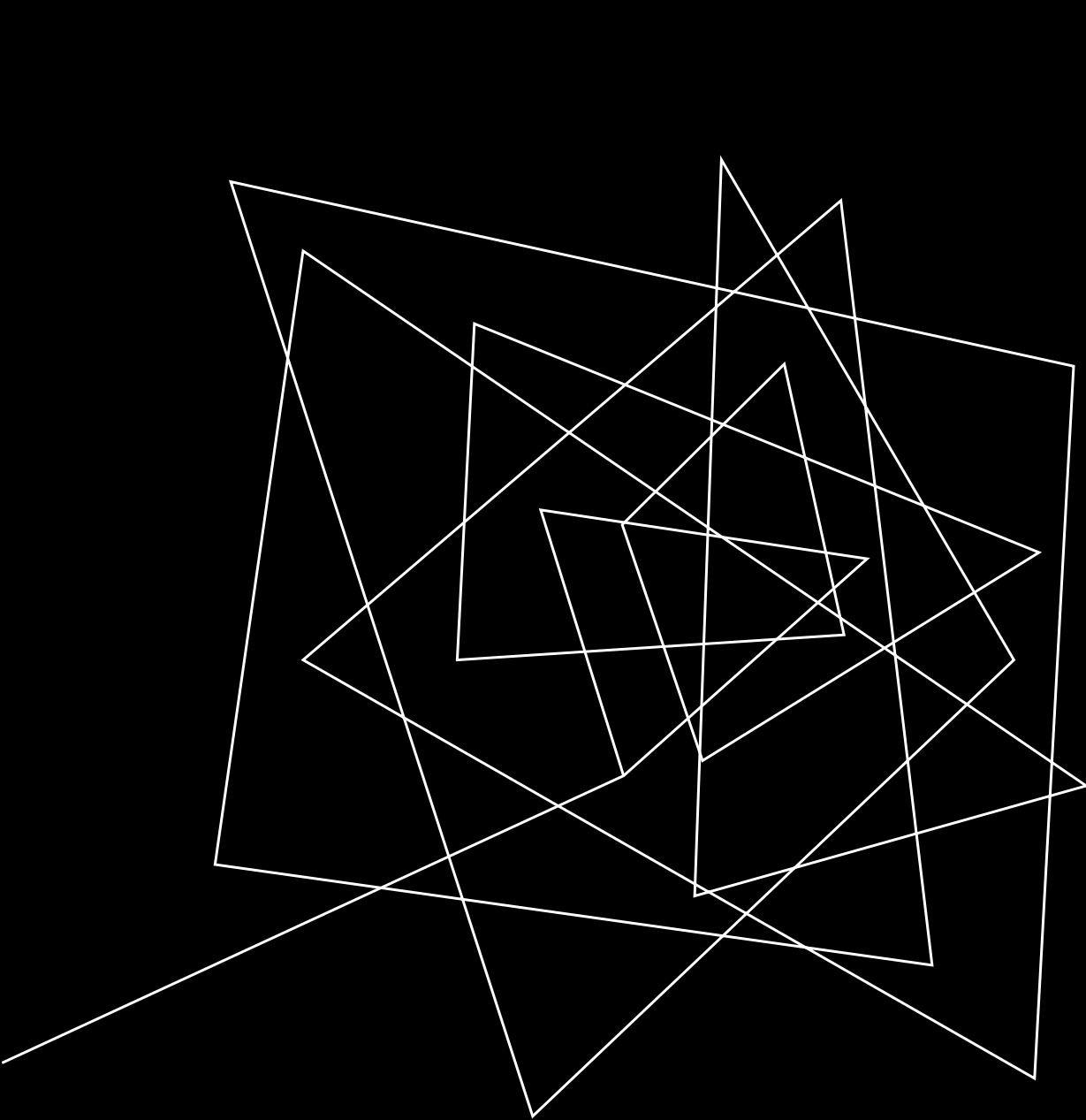
AMS Special Session on Creating, Doing, and Sharing Modeling Approaches to Teaching Differential Equations



https://web.williams.edu/Mathematics/sjmillier/public_html/math/talks/talks.html

Supported in part by NSF Grant number: DMS2341670

Sections

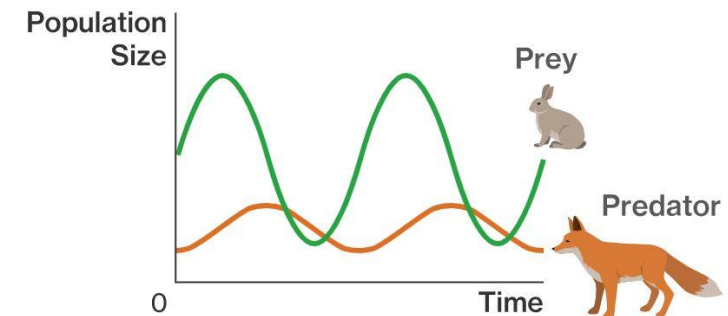
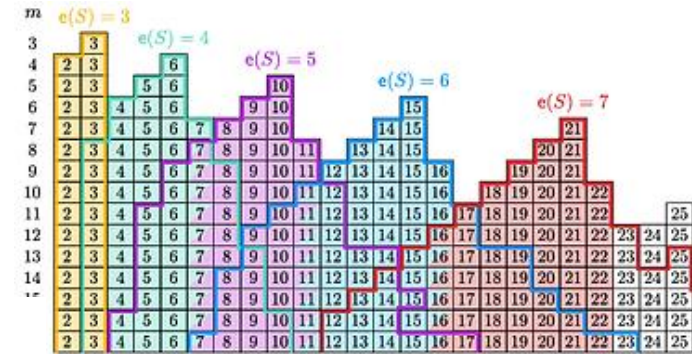
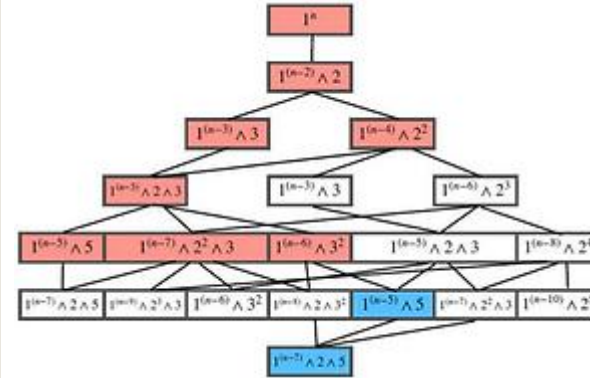
- 
- 1) Polymath Jr.
 - 2) Know your team
 - 3) Research that makes a difference
 - 4) PP models and how to approach
 - 5) Results, Collaborations, future steps, and lessons learned

Polymath Jr.

Our goal is to provide research opportunities to every undergraduate who wishes to explore advanced mathematics. this online program consists of research projects in a variety of mathematical topics and runs in the spirit of the polymath project.

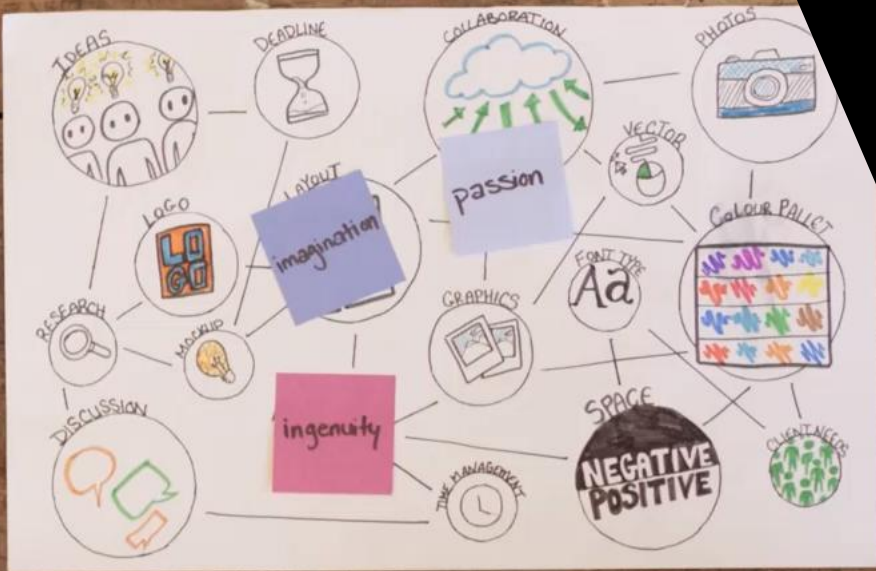
Each project is mentored by an active researcher with experience in undergraduate mentoring.

<https://geometrynyc.wixsite.com/polymathreu>



How Polymath Works

- 1) Polymath accepts hundreds of applications every year, with priority given to seniors.
- 2) All students attend a series of online sessions that introduce a wide range of topics and provide insight into potential projects.
- 3) Mentors receive training on handling stressful situations, leading effectively, giving students space, motivating them, and more.
- 4) Each graduate mentor is paired with a faculty supervisor who guides and oversees the project.
- 5) After these sessions, students choose their top three projects. Students are then matched with a mentor and a supervisor.



Meet the Team



Jasmine Pham



Jessie Wang

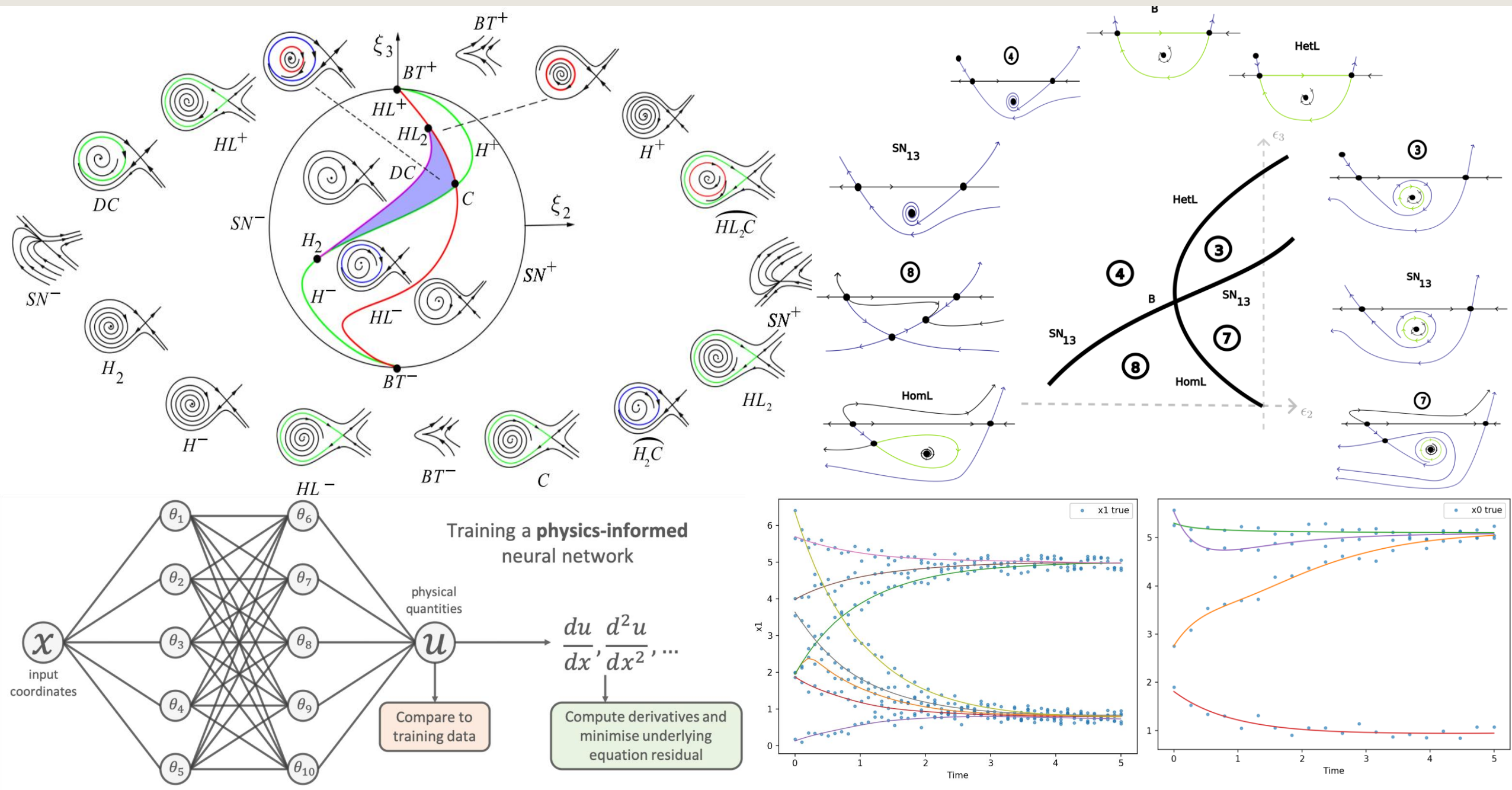


Wael El Khateeb



Steven Miller

My Background



PHASE 1: Introduction

How to guide students without prior differential equations experience to produce high-quality research in just six weeks?

We established a clear framework built on the following principles.

Consistent and Timely Communication

- The average email response time, amazingly, did not exceed 30 minutes.

Restricted Material

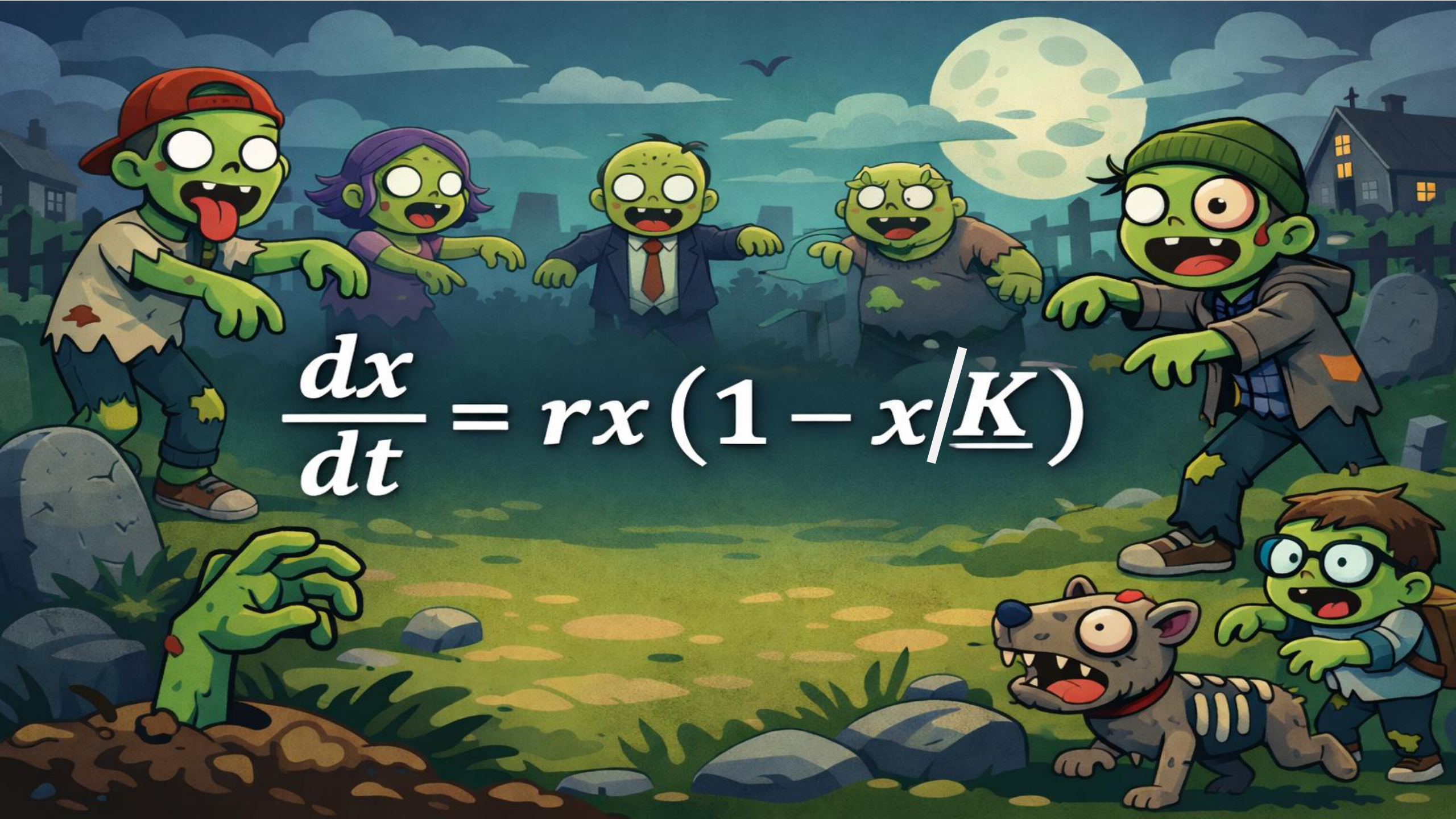
- Restricting our sharing of knowledge to only the essential tools (System of ODEs, Jacobian, basic bifurcations,...).

Weekly group meetings

- Complemented by one-on-one office hours.

Positive Reinforcement and motivation

- Maintain high morale throughout the program to encourage confidence and sustained engagement.

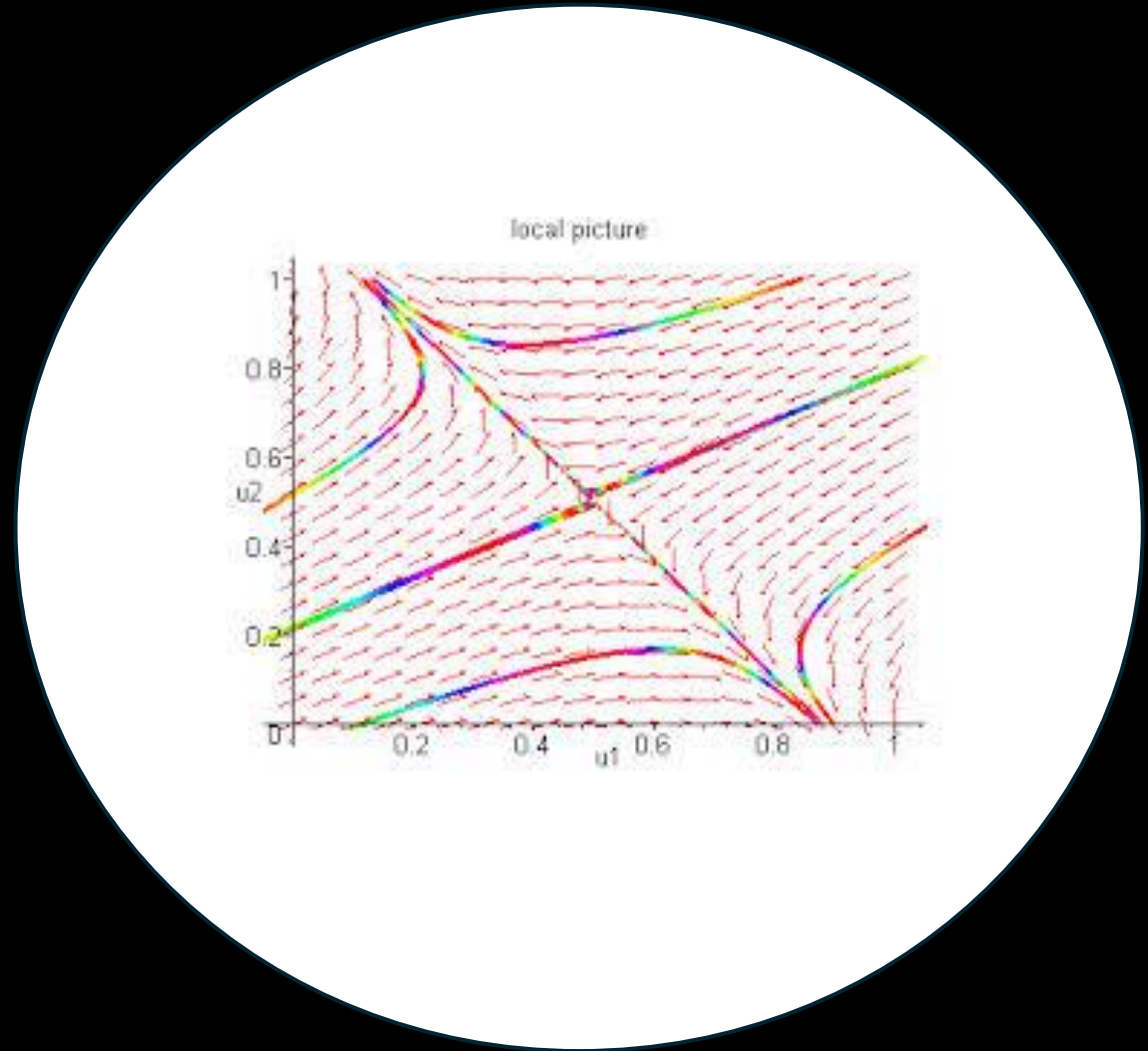


$$\frac{dx}{dt} = rx \left(1 - \frac{x}{K} \right)$$

Find the values of the functions
 $f_1(x) = x, f_2(x) = x^2, f_3(x) = x^3$
near $x = 100, x = 1000, x = 10000$

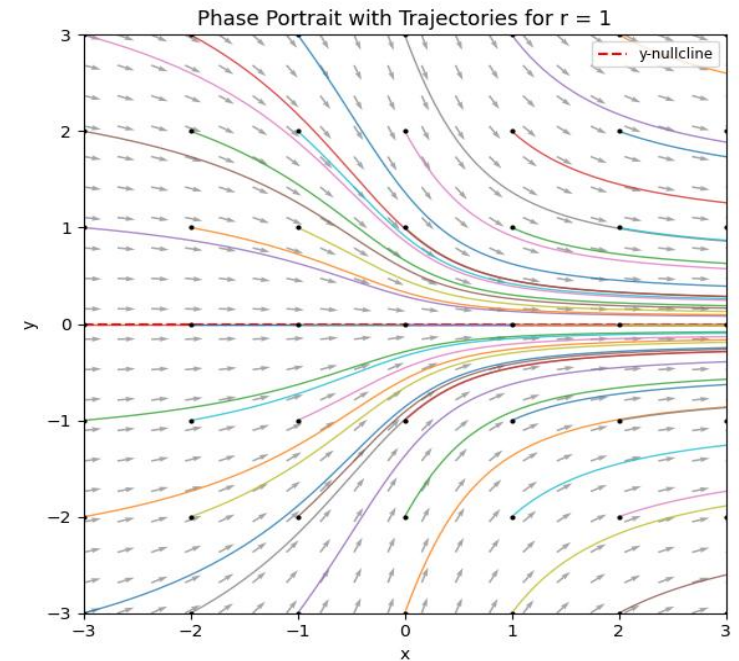
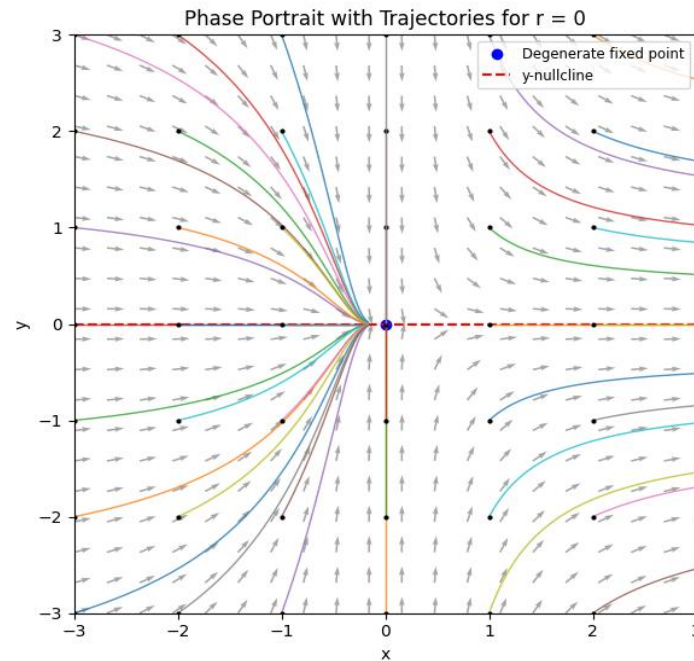
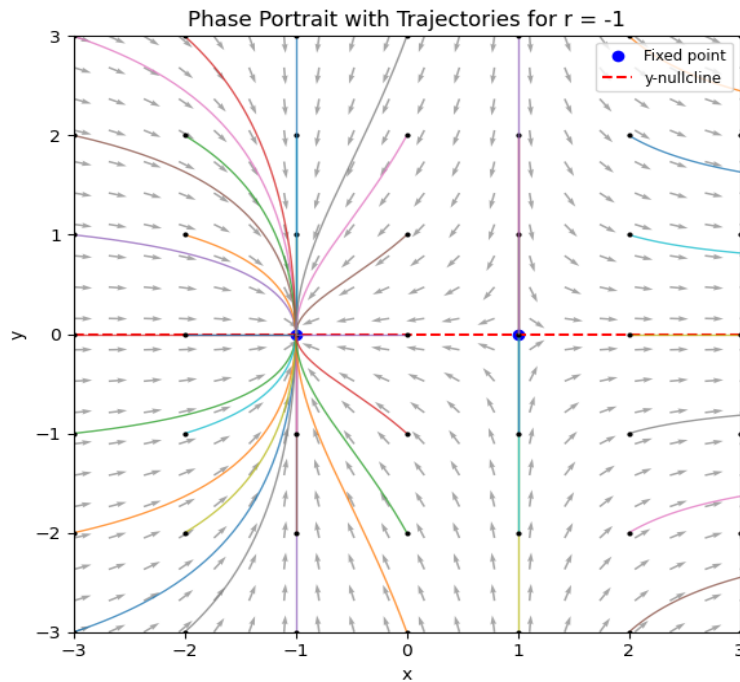
Now do the same for
 $x = 0.01, x = 0.001, x = 0.0001$

What can you conclude?

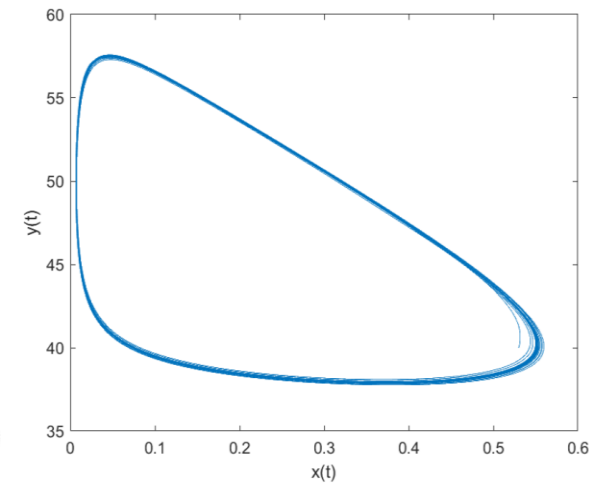
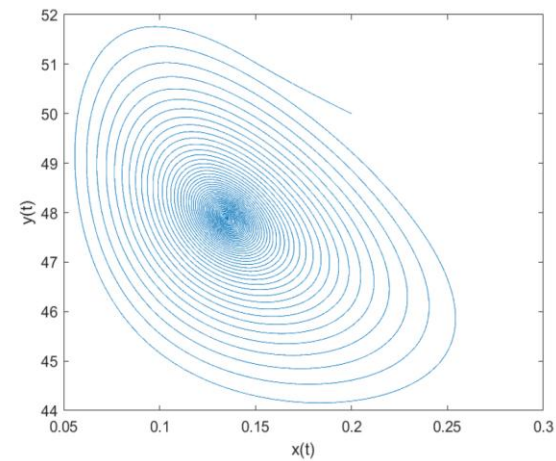
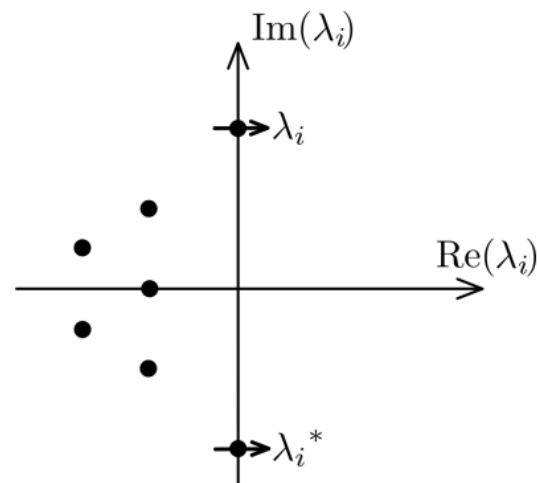


Bifurcations: qualitative change in the behavior of a system as a parameter is varied.

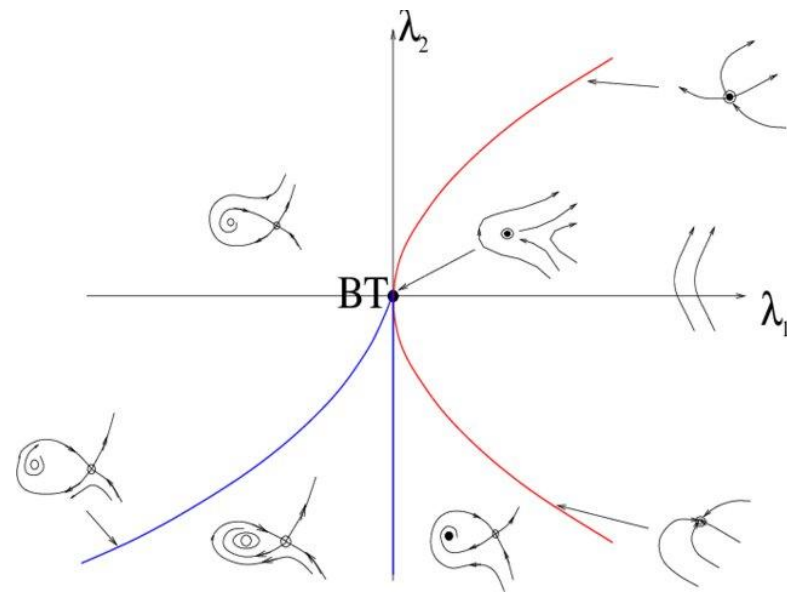
$$\begin{cases} \frac{dx}{dt} = r + x^2, \\ \frac{dy}{dt} = -y. \end{cases}$$



A **Hopf bifurcation** occurs when a fixed point of a dynamical system loses stability as a pair of complex conjugate eigenvalues crosses the imaginary axis, resulting in the emergence or disappearance of a periodic orbit of small amplitude.



What Happens if both Hopf and Saddle node occur simultaneously?



A photograph of an alligator in a swampy environment, partially submerged in water and surrounded by dense vegetation and small white flowers. The image is positioned on the left side of the slide, with a diagonal line separating it from the text area.

BACKGROUND (BIOLOGY)

1800s to mid-1900s: Intense hunting for leather and meat.

1950s to 1960s: Populations crash; poaching persists; weak enforcement.

1967: Listed as endangered in the United States.

1973: Endangered Species Act; stronger enforcement; interstate hide trade curbed.

1970s: States close hunting, protect nests, begin science-based management.

BACKGROUND (BIOLOGY)

Late 1970s to 1980s: Rebound via protection, anti-poaching, wetland conservation.

1987: Delisted federally; managed as “threatened due to similarity of appearance.”

Today: Millions across the Southeast, especially Florida and Louisiana.

FWC lottery permits; two alligators per permit.



Burmese Pythons

Invasive to the everglades,
they were released by
means of



- Intentional releases by owners who could no longer keep large snakes.
- Accidental escapes from pet stores, breeders, and private collections.
- Facility damage during Hurricane Andrew (1992) likely released additional snakes into the Everglades.

ECOLOGICAL IMPLICATIONS

- Sharp declines of midsized mammals in parts of the Everglades (raccoons, opossums, marsh rabbits)
- Predation on native birds, reptiles, and occasional juvenile alligators
- Pressure on threatened species that nest or forage on the ground
- Competition with native apex predators for prey, altering food webs

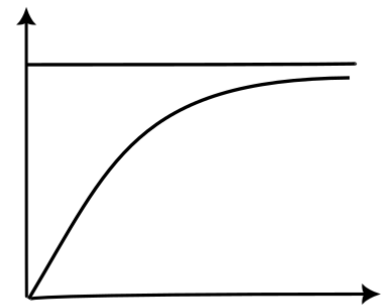


PP model and extensions

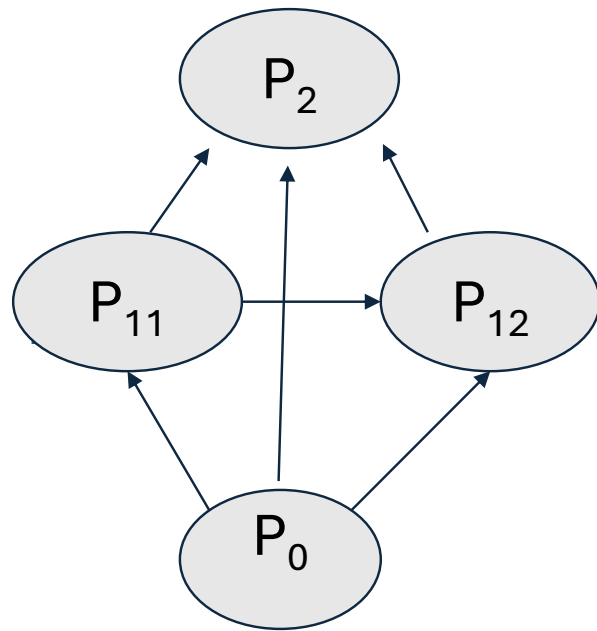


$$\frac{dx}{dt} = G_1(x) - \alpha_1 F(x)y - H_1(x)$$

$$\frac{dy}{dt} = -G_2(y) + \alpha_2 F(x)y - H_2(y)$$



Model Illustrated:



P_0 - Raccoon/prey population

P_{11} - Juvenile Alligator population

P_{12} - Adult Alligator population

P_2 - Burmese Python population

Initial Model

$$f_m(P_n) = P_n$$

$$H_2(P_2) = h_2$$

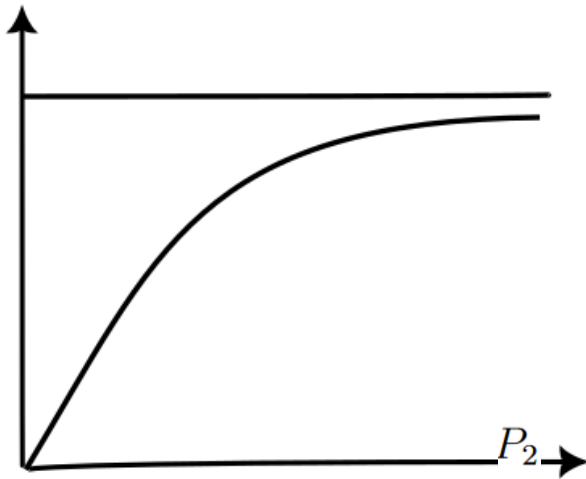
$$H_3(P_3) = h_3 P_3$$

$$\begin{cases} \frac{dP_0}{dt} = f_1(P_0) - m_{20}f_2(P_0)P_{12} - m_{30}f_3(P_0)P_2 \\ \frac{dP_{11}}{dt} = c_{20}f_2(P_0)P_{12} - m_{31}f_4(P_{11})P_3 - d_1 P_{11} \\ \frac{dP_{12}}{dt} = d_1 P_{11} - d_2 P_{12} - H_2(P_{12}) \\ \frac{dP_2}{dt} = c_{30}f_3(P_0)P_2 + c_{31}f_4(P_{11})P_2 - d_3 P_2 - H_3(P_2) \end{cases}$$

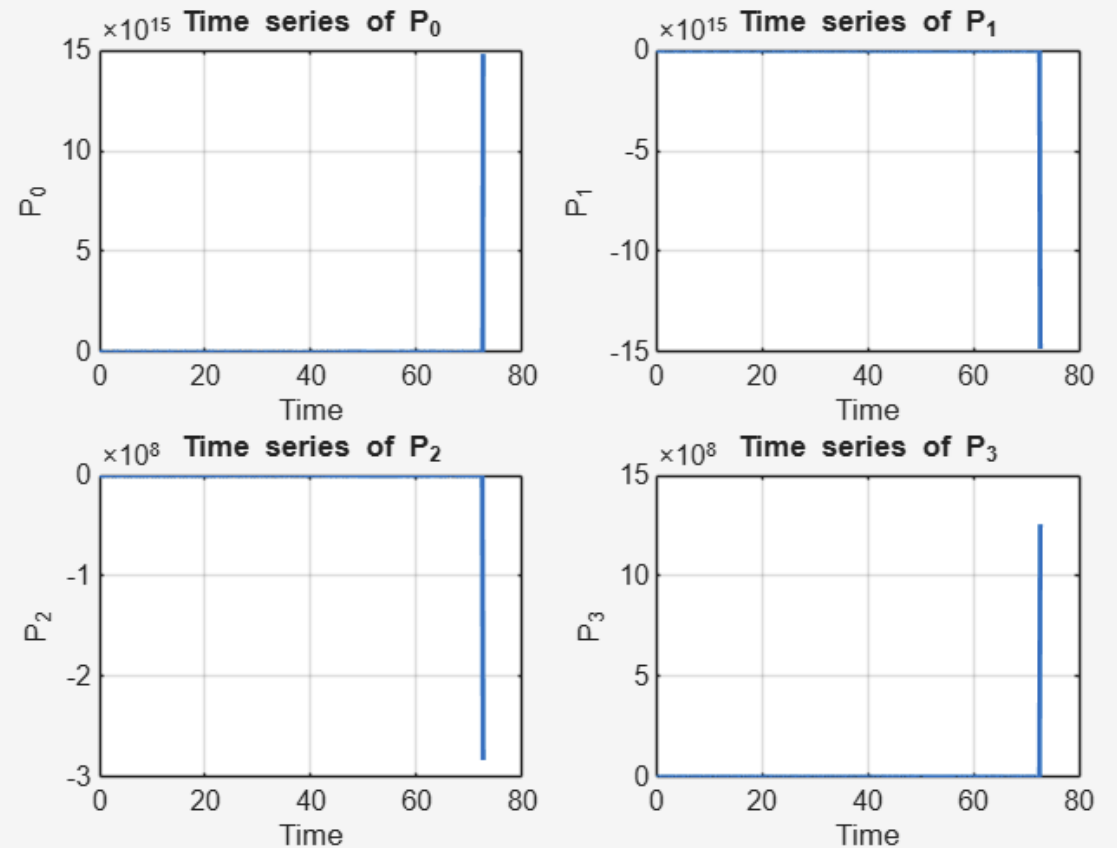
$$\begin{cases} \frac{dP_0}{dt} = r P_0 - m_{20} P_0 P_{12} - m_{30} P_0 P_2 \\ \frac{dP_{11}}{dt} = c_{20} P_0 P_{12} - m_{31} P_{11} P_3 - d_{11} P_{11} \\ \frac{dP_{12}}{dt} = d_{12} P_{11} - d_2 P_{12} - h_2 \\ \frac{dP_2}{dt} = c_{30} P_0 P_2 + c_{31} P_2 P_{11} - d_3 P_2 - h_3 P_2 \end{cases} \xrightarrow{h_2 \Rightarrow \text{Saturated}} \begin{cases} \frac{dP_0}{dt} = r P_0 - m_{20} P_0 P_{12} - m_{30} P_0 P_2 \\ \frac{dP_{11}}{dt} = c_{20} P_0 P_{12} - m_{31} P_{11} P_3 - d_{11} P_{11} \\ \frac{dP_{12}}{dt} = d_{12} P_{11} - d_2 P_{12} - h_2 \frac{P_{12}}{a + P_{12}} \\ \frac{dP_2}{dt} = c_{30} P_0 P_2 + c_{31} P_2 P_{11} - d_3 P_2 - h_3 P_2 \end{cases}$$

Why constant harvesting doesn't work

Applying constant harvesting shows that P_1 and P_2 eventually become **negative**, which is biologically unrealistic and highlights why **constant harvesting does not work**.



Simulation of 4-ODE system (P2 harvesting: constant)



Final Model

Prey	$\left\{ \begin{array}{l} \frac{dP_0}{dt} = rP_0 - m_{20}P_0P_{12} - m_{30}P_0P_2 \\ \frac{dP_{11}}{dt} = c_{20}P_0P_{12} - m_{31}P_{11}P_3 - d_{11}P_{11} \\ \frac{dP_{12}}{dt} = d_{12}P_{11} - P_{12}P_2 - h_2 \frac{P_{12}}{a + P_{12}} \\ \frac{dP_2}{dt} = c_{30}P_0P_2 + c_{31}P_2P_{11} - d_3P_2 - h_3P_2 \end{array} \right.$
Juvenile Alligator	
Adult Alligator	
Python	



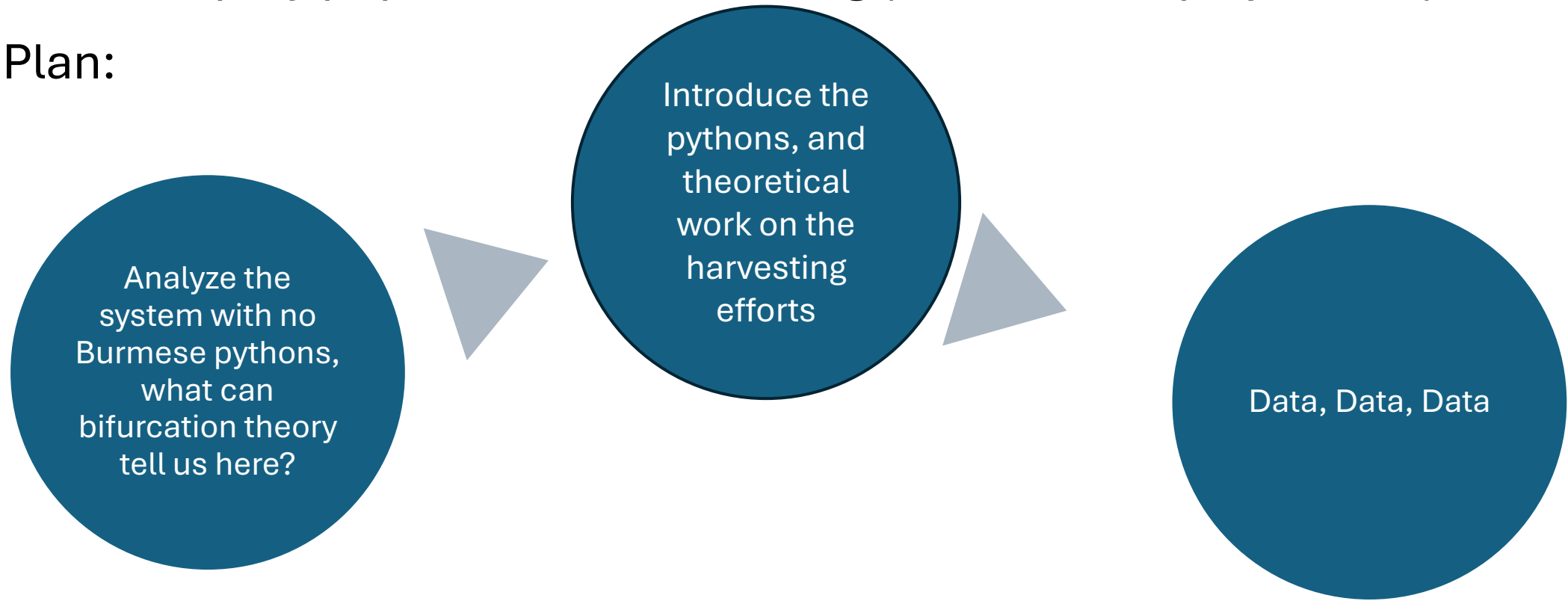
$$\left\{ \begin{array}{l} \frac{dP_0}{dt} = rP_0 \left(1 - \frac{P_0}{K} \right) - m_{10}P_0P_1 - m_{20}P_0P_2, \\ \frac{dP_1}{dt} = c_{10}P_0P_1 - d_1P_1 - \frac{h_1P_1}{a + P_1} - m_{31}P_1P_2, \\ \frac{dP_2}{dt} = c_{20}P_0P_2 + c_{21}P_1P_2 - d_2P_2 - h_2P_2. \end{array} \right.$$

Equilibria: 7 in total

- 2 Equilibria on the P_0 axis.
- 2 equilibria on the $P_0 - P_1$ plane suggesting that the prey can coexist with the alligators.
- 2 non-zero equilibria where all 3 species coexist.
- 1 equilibrium in the $P_0 - P_2$ plane suggesting that python can coexist with the general prey.

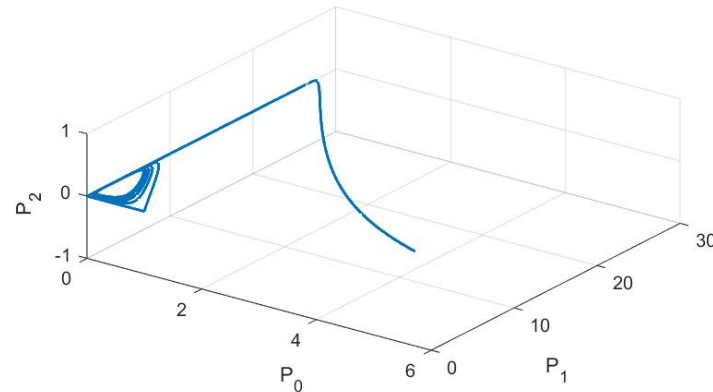
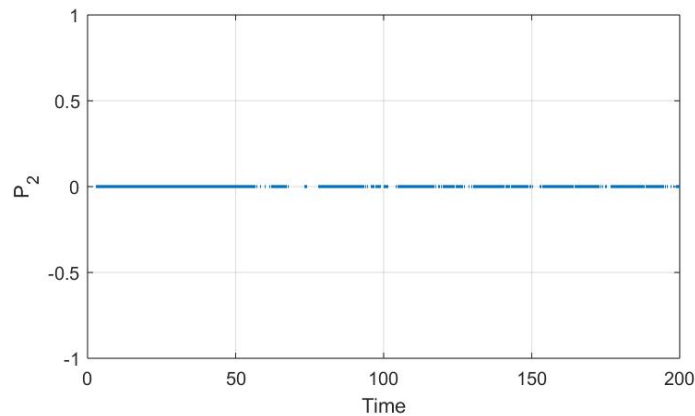
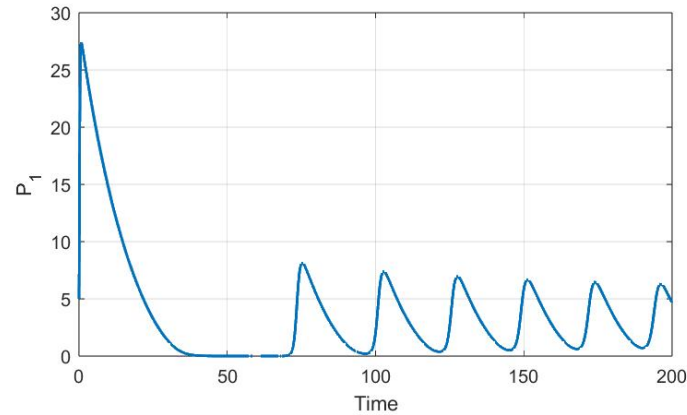
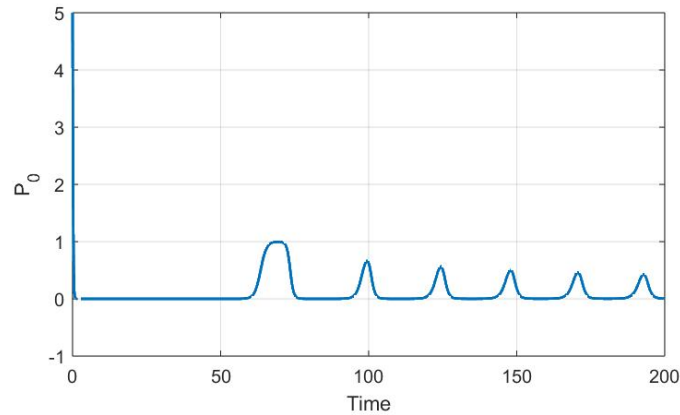
Ultimate goal: Policy making

- Our model predicts that Alligators and prey can coexist, as well as general prey and Burmese pythons can also coexist. We can also see that all three species can coexist
- With current harvesting restricted to 2 alligators per hunter, per season, should the hunting policies be altered, especially that the common prey population is decreasing (Racoons drop by 95%...)
- Plan:



THE CASE WITH NO BURMESE PYTHONS

- If no Burmese Pythons are present our 3D model is reduced to a 2D system, the system hereby would exhibit 2 non-zero equilibria: Call them E_1 and E_2 .



Steps followed:

1. **Shift the equilibrium:** Find the equilibrium $P^* = (P_0^*, P_1^*, P_2^*)$ and shift variables:

$$u = P_0 - P_0^*, \quad v = P_1 - P_1^*, \quad w = P_2 - P_2^*$$

so that the system becomes $\dot{z} = F(z)$ with $F(0) = 0$.

2. **Linearize and identify eigen-directions:** Compute the Jacobian $J = DF(0)$. At the BT point: $\text{spec}(J) = \{0, 0, \lambda_s\}$ with $\lambda_s < 0$. The eigenvectors corresponding to zero eigenvalues define the center subspace.

3. **Center manifold reduction (3D $\rightarrow$ 2D):** There exists a smooth invariant manifold $w = h(u, v)$ with $h(0) = 0$, $Dh(0) = 0$. Substituting $w = h(u, v)$ reduces the system to:

$$\dot{u} = \tilde{F}_1(u, v), \quad \dot{v} = \tilde{F}_2(u, v)$$

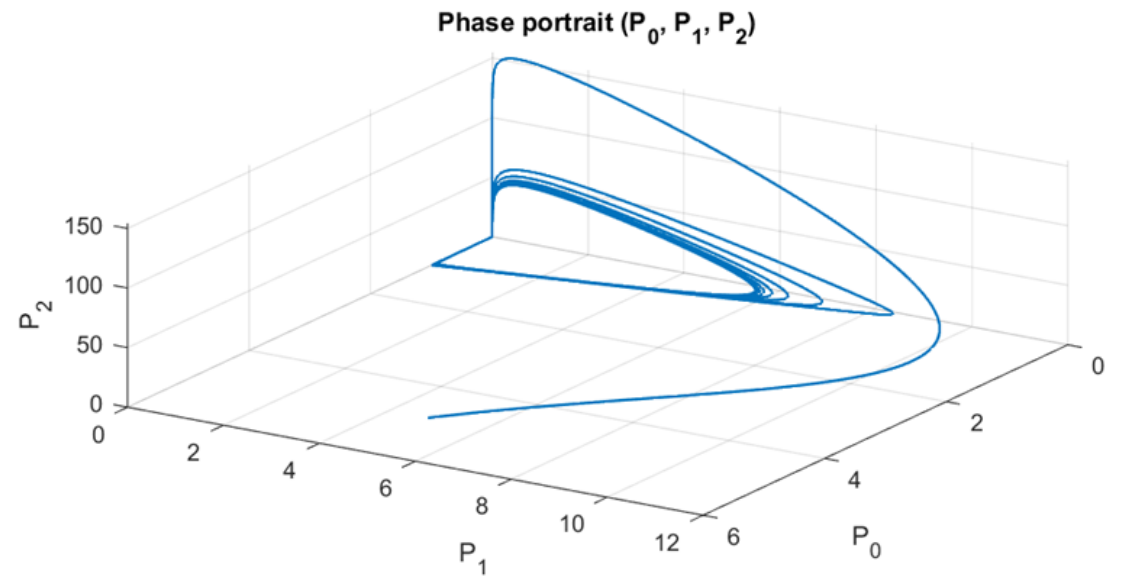
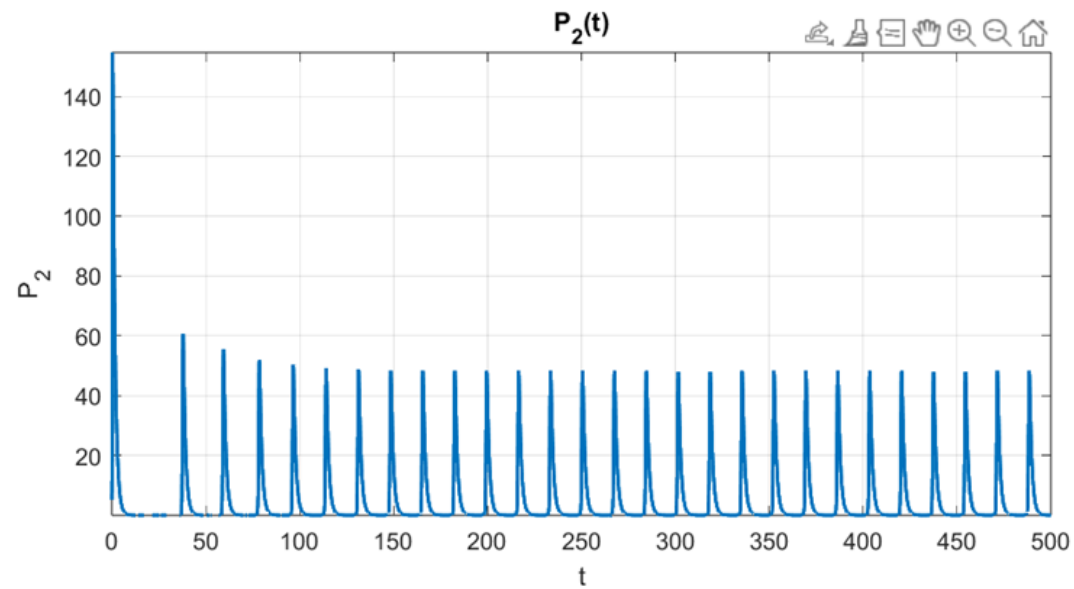
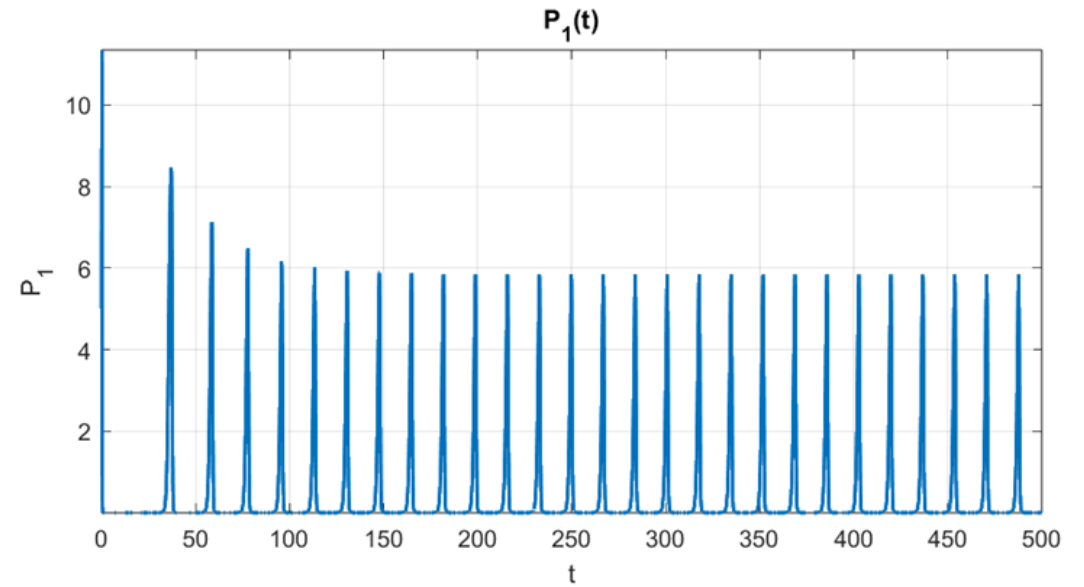
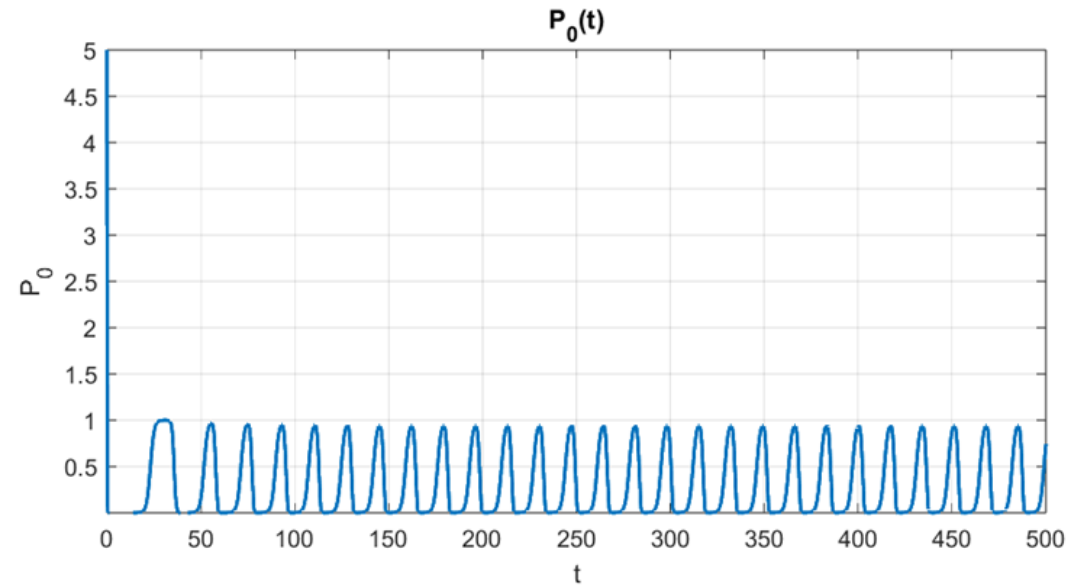
4. **Reduced dynamics:** Expand $\tilde{F}_1, \tilde{F}_2$ in a Taylor series near the origin and retain quadratic terms:

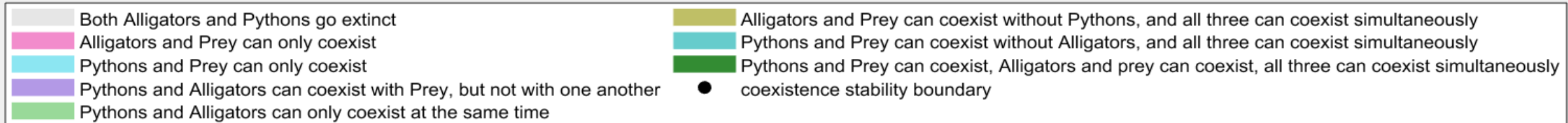
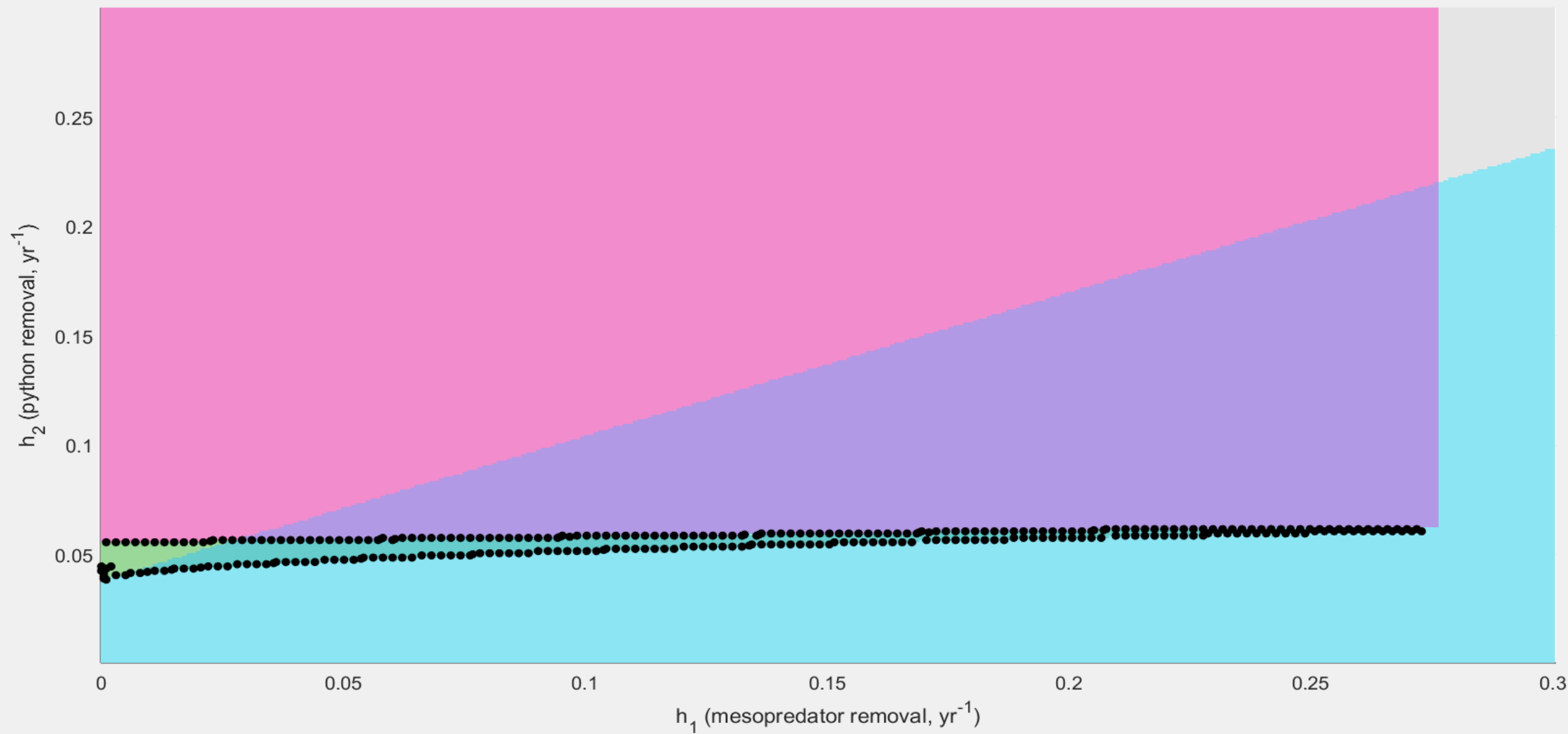
$$\dot{u} = v + \mathcal{O}(2), \quad \dot{v} = au^2 + buv + \dots$$

5. **Transform to BT normal form:** Apply smooth coordinate and parameter changes to obtain:

$$\boxed{\dot{x} = y, \quad \dot{y} = \beta_1 + \beta_2 x + ax^2 + bxy + \mathcal{O}(3)}$$

The presence of Burmese Pythons





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Useful techniques/ Strategies implemented

- **Ownership**
- **Think-Pair-Share**
- **Know your strong points, know your weak points**
- **Always have something to share**



Thank you!

Steven J Miller
Professor
Williams College



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Ph.D. Candidate in Applied Math
The University of Toledo



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